

15

CHAPTER



Employee Stock Options

Employee stock options are call options on a company's stock granted by the company to its employees. The options give the employees a stake in the fortunes of the company. If the company does well so that the company's stock price moves above the strike price, employees gain by exercising the options and then selling the stock they acquire at the market price.

Employee stock options have become very popular in the last 20 years. Many companies, particularly technology companies, feel that the only way they can attract and keep the best employees is to offer them very attractive stock option packages. Some companies grant options only to senior management; others grant them to people at all levels in the organization. Microsoft was one of the first companies to use employee stock options. All Microsoft employees were granted options and, as the company's stock price rose, it is estimated that over 10,000 of them became millionaires. In 2003 Microsoft announced that it would discontinue the use of options and award shares of Microsoft to employees instead. But many other companies throughout the world continue to be enthusiastic users of employee stock options.

Employee stock options are popular with start-up companies. Often these companies do not have the resources to pay key employees as much as they could earn with an established company and they solve this problem by supplementing the salaries of the employees with stock options. If the company does well and shares are sold to the public in an IPO, the options are likely to prove to be very valuable. Some newly formed companies have even granted options to students who worked for just a few months during their summer break—and in some cases this has led to windfalls of hundreds of thousands of dollars for the students!

This chapter explains how stock option plans work and how their popularity has been influenced by their accounting treatment. It discusses whether employee stock options help to align the interests of shareholders with those of top executives running a company. It also describes how these options are valued and looks at backdating scandals.

15.1 CONTRACTUAL ARRANGEMENTS

Employee stock options often last as long as 10 to 15 years. Very often the strike price is set equal to the stock price on the grant date so that the option is initially at the money.

The following are usually features employee stock option plans:

1. There is a vesting period during which the options cannot be exercised. This vesting period can be as long as four years.
2. When employees leave their jobs (voluntarily or involuntarily) during the vesting period, they forfeit their options.
3. When employees leave (voluntarily or involuntarily) after the vesting period, they forfeit options that are out of the money and they have to exercise vested options that are in the money almost immediately.
4. Employees are not permitted to sell the options.
5. When an employee exercises options, the company issues new shares and sells them to the employee for the strike price.

The Early Exercise Decision

The fourth feature of employee stock option plans noted above has important implications. If employees, for whatever reason, want to realize a cash benefit from options that have vested, they must exercise the options and sell the underlying shares. They cannot sell the options to someone else. This leads to a tendency for employee stock options to be exercised earlier than similar regular exchange-traded or over-the-counter call options.

Consider a call option on a stock paying no dividends. In Section 10.5 we showed that, if it is a regular call option, it should never be exercised early. The holder of the option will always do better by selling the option rather than exercising it before the end of its life. However, the arguments we used in Section 10.5 are not applicable to employee stock options because they cannot be sold. The only way employees can realize a cash benefit from the options (or diversify their holdings) is by exercising the options and selling the stock. It is therefore not unusual for an employee stock option to be exercised well before it would be optimal to exercise the option if it were a regular exchange-traded or over-the-counter option.

Should an employee ever exercise his or her options before maturity and then keep the stock rather than selling it? Assume that the option's strike price is constant during the life of the option and the option can be exercised at any time. To answer the question we consider two options: the employee stock option and an otherwise identical regular option that can be sold in the market. We refer to the first option as option A and the second as option B. If the stock pays no dividends, we know that option B should never be exercised early. It follows that it is not optimal to exercise option A and keep the stock. If the employee wants to maintain a stake in his or her company, a better strategy is to keep the option. This delays paying the strike price and maintains the insurance value of the option, as described in Section 10.5. Only when it is optimal to exercise option B can it be a rational strategy for an employee to exercise option A before maturity and keep the stock.¹ As discussed in Section 14.12, it is optimal to exercise option B only when a relatively high dividend is imminent.

In practice the early exercise behavior of employees varies widely from company to company. In some companies, there is a culture of not exercising early; in others, employees tend to exercise options and sell the stock soon after the end of the vesting period, even if the options are only slightly in the money.

¹ The only exception to this could be when an executive wants to own the stock for its voting rights.

15.2 DO OPTIONS ALIGN THE INTERESTS OF SHAREHOLDERS AND MANAGERS?

For investors to have confidence in capital markets, it is important that the interests of shareholders and managers are reasonably well aligned. This means that managers should be motivated to make decisions that are in the best interests of shareholders. Managers are the agents of the shareholders and, as mentioned in Chapter 8, economists use the term *agency costs* to describe the losses experienced when the interests of agents and principals are not aligned.

Do employee stock options help align the interests of employees and shareholders? The answer to this question is not straightforward. There can be little doubt that they serve a useful purpose for a start-up company. The options are an excellent way for the main shareholders, who are usually also senior executives, to motivate employees to work long hours. If the company is successful and there is an IPO, the employees will do very well; but if the company is unsuccessful, the options will be worthless.

It is the options granted to the senior executives of publicly traded companies that are most controversial. It has been estimated that employee stock options account for about 50% of the remuneration of top executives in the United States. Executive stock options are sometimes referred to as an executive's "pay for performance." If the company's stock price goes up, so that shareholders make gains, the executive is rewarded. However, this overlooks the asymmetric payoffs of options. If the company does badly then the shareholders lose money, but all that happens to the executives is that they fail to make a gain. Unlike the shareholders, they do not experience a loss.² A better type of pay for performance involves the simpler strategy of giving stock to executives. The gains and losses of the executives then mirror those of other shareholders.

What temptations do stock options create for a senior executive? Suppose an executive plans to exercise a large number of stock options in three months and sell the stock. He or she might be tempted to time announcements of good news—or even move earnings from one quarter to another—so that the stock price increases just before the options are exercised. Alternatively, if at-the-money options are due to be granted to the executive in three months, the executive might be tempted to take actions that reduce the stock price just before the grant date. The type of behavior we are talking about here is of course totally unacceptable—and may well be illegal. But the backdating scandals, which are discussed later in this chapter, show that the way some executives have handled issues related to stock options leaves much to be desired.

Even when there is no impropriety of the type we have just mentioned, executive stock options are liable to have the effect of motivating executives to focus on short-term profits at the expense of longer-term performance. In some cases they might even take risks they would not otherwise take (and risks that are not in the interests of the shareholders) because of the asymmetric payoffs of options. Managers of large funds worry that, because stock options are such a huge component of an executive's compensation, they are liable to be a big source of distraction. Senior management may spend too much time thinking about all the different aspects of their compensation and not enough time running the company!

² When options have moved out of the money, companies have sometimes replaced them with new at-the-money options. This practice known as "repricing" leads to the executive's gains and losses being even less closely tied to those of the shareholders.

A manager's inside knowledge and ability to affect outcomes and announcements is always liable to interact with his or her trading in a way that is to the disadvantage of other shareholders. One radical suggestion for mitigating this problem is to require executives to give notice to the market—perhaps one week's notice—of an intention to buy or sell their company's stock.³ (Once the notice of an intention to trade had been given, it would be binding on the executive.) This allows the market to form its own conclusions about why the executive is trading. As a result, the price may increase before the executive buys and decrease before the executive sells.

15.3 ACCOUNTING ISSUES

An employee stock option represents a cost to the company and a benefit to the employee just like any other form of compensation. This point, which for many is self-evident, is actually quite controversial. Many corporate executives appear to believe that an option has no value unless it is in the money. As a result, they argue that an at-the-money option issued by the company is not a cost to the company. The reality is that, if options are valuable to employees, they must represent a cost to the company's shareholders—and therefore to the company. There is no free lunch. The cost to the company of the options arises from the fact that the company has agreed that, if its stock does well, it will sell shares to employees at a price less than that which would apply in the open market.

Prior to 1995 the cost charged to the income statement of a company when it issued stock options was the intrinsic value. Most options were at the money when they were first issued, so that this cost was zero. In 1995, accounting standard FAS 123 was issued. Many people expected it to require the expensing of options at their fair value. However, as a result of intense lobbying, the 1995 version of FAS 123 only encouraged companies to expense the fair value of the options they granted on the income statement. It did not require them to do so. If fair value was not expensed on the income statement, it had to be reported in a footnote to the company's accounts.

Accounting standards have now changed to require the expensing of stock options at their fair value on the income statement. In February 2004 the International Accounting Standards Board issued IAS 2 requiring companies to start expensing stock options in 2005. In December 2004 FAS 123 was revised to require the expensing of employee stock options in the United States starting in 2005.

The effect of the new accounting standards is to require options to be valued on the grant date and the amount to be recorded as an expense in the income statement for the year in which the grant is made. Valuation at a time later than the grant date is not required. It can be argued that options should be revalued at financial year ends (or every quarter) until they are exercised or reach the end of their lives.⁴ This would treat them in the same way as other derivative transactions entered into by the company. If the option became more valuable from one year to the next, there would then be an

³ This would apply to the exercise of options because, if an executive wants to exercise options and sell the stock that is acquired, then he or she would have to give notice of intention to sell.

⁴ See J. Hull and A. White, "Accounting for Employee Stock Options: A Practical Approach to Handling the Valuation Issues," *Journal of Derivatives Accounting*, 1, 1 (2004): 3–9.

additional amount to be expensed. However, if it declined in value, there would be a positive impact on income.

This approach would have a number of advantages. The cumulative charge to the company would reflect the actual cost of the options (either zero if the options are not exercised or the option payoff if they are exercised). Although the charge in any year would depend on the option pricing model used, the cumulative charge over the life of the option would not.⁵ Arguably there would be much less incentive for the company to engage in the backdating practices described later in the chapter. The disadvantage usually cited for accounting in this way is that it is undesirable because it introduces volatility into the income statement.⁶

Nontraditional Option Plans

It is easy to understand why pre-2005 employee stock options tended to be at the money on the grant date and have strike prices that did not change during the life of the option. Any departure from this standard arrangement was likely to require the options to be expensed. Now that accounting rules have changed so that all options are expensed at fair value, many companies are considering alternatives to the standard arrangement.

One argument against the standard arrangement is that employees do well when the stock market goes up, even if their own company's stock price does less well than the market. One way of overcoming this problem is to tie the strike price of the options to the performance of the S&P 500. Suppose that on the option grant date the stock price is \$30 and the S&P 500 is 1,500. The strike price would initially be set at \$30. If the S&P 500 increased by 10% to 1,650, then the strike price would also increase by 10% to \$33. If the S&P 500 moved down by 15% to 1,275, then the strike price would also move down by 15% to \$25.50. The effect of this is that the company's stock price performance has to beat the performance of the S&P 500 to become in the money. As an alternative to using the S&P 500 as the reference index, the company could use an index of the prices of stocks in the same industrial sector as the company.

In another variation on the standard arrangement, the strike price increases through time in a predetermined way such that the shares of the stock have to provide a certain minimum return per year for the options to be in the money. In some cases, profit targets are specified and the options vest only if the profit targets are met.⁷

15.4 VALUATION

Accounting standards give companies quite a bit of latitude in choosing a method for valuing employee stock options. In this section we review some of the alternatives.

⁵ Interestingly, if an option is settled in cash rather than by the company issuing new shares, it is subject to the accounting treatment proposed here. (However, there is no economic difference between an option that is settled in cash and one that is settled by selling new shares to the employee.)

⁶ In fact the income statement is likely be less volatile if stock options are revalued. When the company does well, income is reduced by revaluing the executive stock options. When the company does badly, it is increased.

⁷ This type of option is difficult to value because the payoff depends on reported accounting numbers as well as the stock price. Valuations usually assume that targets will be met.

The “Quick and Dirty” Approach

A frequently used approach is based on what is known as the option’s *expected life*. This is the average time for which employees hold the option before it is exercised or expires. The expected life can be approximately estimated from historical data on the early exercise behavior of employees and reflects the vesting period, the impact of employees leaving the company, and the tendency we mentioned in Section 15.1 for employee stock options to be exercised earlier than regular options. The Black–Scholes–Merton model is used with the life of the option, T , set equal to the expected life. The volatility is usually estimated from several years of historical data as described in Section 14.4.

It should be emphasized that using the Black–Scholes–Merton formula in this way has no theoretical validity. There is no reason why the value of a European stock option with the time to maturity, T , set equal to the expected life should be approximately the same as the value of the American-style employee stock option that we are interested in. However, the results given by the model are not totally unreasonable. Companies, when reporting their employee stock option expense, will frequently mention the volatility and expected life used in their Black–Scholes–Merton computations.

Example 15.1

A company grants 1,000,000 options to its executives on November 1, 2011. The stock price on that date is \$30 and the strike price of the options is also \$30. The options last for 10 years and vest after three years. The company has issued similar at-the-money options for the last 10 years. The average time to exercise or expiry of these options is 4.5 years. The company therefore decides to use an “expected life” of 4.5 years. It estimates the long-term volatility of the stock price, using 5 years of historical data, to be 25%. The present value of dividends during the next 4.5 years is estimated to be \$4. The 4.5-year zero-coupon risk-free interest rate is 5%. The option is therefore valued using the Black–Scholes–Merton model (adjusted for dividends in the way described in Section 14.12) with $S_0 = 30 - 4 = 26$, $K = 30$, $r = 5\%$, $\sigma = 25\%$, and $T = 4.5$. The Black–Scholes–Merton formula gives the value of one option as \$6.31. Hence, the income statement expense is $1,000,000 \times 6.31$, or \$6,310,000.

Binomial Tree Approach

A more sophisticated approach to valuing employee stock options involves building a binomial tree as outlined in Chapter 12 and adjusting the rules used when rolling back through the tree to reflect (a) whether the option has vested, (b) the probability of the employee leaving the company, and (c) the probability of the employee choosing to exercise the option. The terms of the option define whether the option has vested at different nodes of the tree. Historical data on turnover rates for employees can be used to estimate the probability of the option being either prematurely exercised or forfeited at a node because the employee leaves the company. The probability of an employee choosing to exercise the option at different nodes of the tree is more difficult to quantify. Clearly this probability increases as the ratio of the stock price to the strike price increases and as the time to the option’s maturity declines. If enough historical data is available, the probability of exercise as a function of these two variables can be estimated—at least approximately.

Example 15.2

Suppose a company grants stock options that last 8 years and vest after 3 years. The stock price and strike price are both \$40. The stock price volatility is 30%, the risk-free rate is 5%, and the company pays no dividends. Figure 15.1 shows how a four-step tree could be used to value the option. (This is for illustration; in practice more time steps would be used.) In this case, $\sigma = 0.3$, $\Delta t = 2$, and $r = 0.05$, so that, with the notation of Chapter 12, $a = e^{0.05 \times 2} = 1.1052$, $u = e^{0.3\sqrt{2}} = 1.5285$, $d = 1/u = 0.6543$, and $p = (a - d)/(u - d) = 0.5158$. The probability on the "up branches" is 0.5158 and the probability on the "down branches" is 0.4842. There are three nodes where early exercise could be desirable: D, G, and H. (The option has not vested at node B and is not in the money at the other nodes prior to maturity.) We assume that the probabilities that the holder will choose to exercise at nodes D, G, and H (conditional on no earlier exercise) have been estimated as 40%, 80%, and 30%, respectively. We suppose that the probability of an employee leaving the company during each time step is 5%. (This corresponds to an employee turnover rate of approximately 2.5% per year.) For the purposes of the calculation, it is assumed that employees always leave at the end of a time period. If an employee leaves the company before an option has vested or when the option is out of the money, the option is forfeited. In other cases the option must be exercised immediately.

Figure 15.1 Valuation of employee stock option in Example 15.2.

At each node:

Upper value = Underlying asset price

Lower value = Option price

Values in bold are a result of early exercise.

Strike price = 40

Discount factor per step = 0.9048

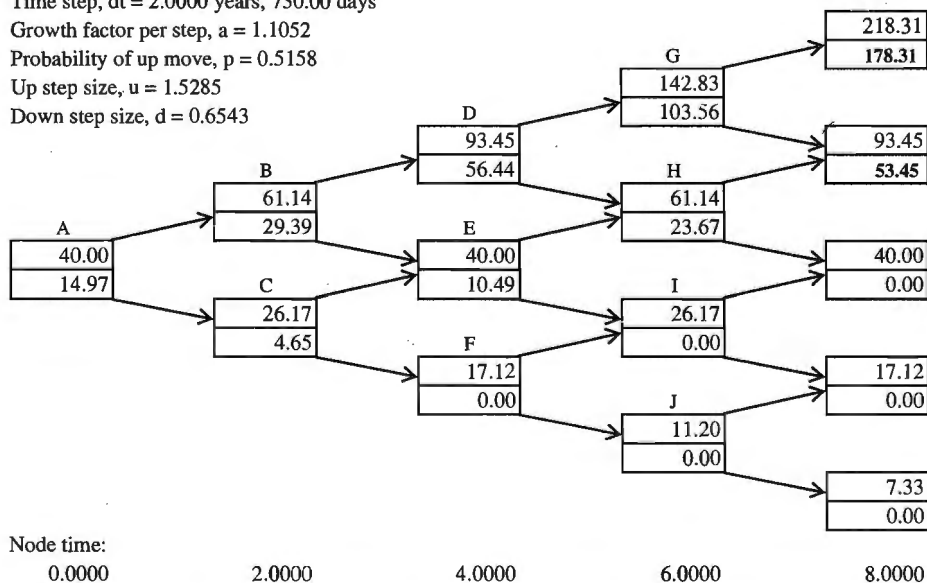
Time step, $dt = 2.0000$ years, 730.00 days

Growth factor per step, $a = 1.1052$

Probability of up move, $p = 0.5158$

Up step size, $u = 1.5285$

Down step size, $d = 0.6543$



The value of the option at the final nodes is its intrinsic value. Consider the nodes at time 6 years. Nodes I and J are easy. Since these nodes are certain to lead to nodes where the option is worth nothing, the value of the option is zero at these nodes. At node H there is a 30% chance that the employee will choose to exercise the option. In cases where the employee does not choose to exercise, there is a 5% chance that the employee leaves the company and has to exercise. The total probability of exercise is therefore $0.3 + 0.7 \times 0.05 = 0.335$. If the option is exercised, its value is $61.14 - 40 = 21.14$. If it is not exercised, its value is

$$e^{-0.05 \times 2}(0.5158 \times 53.45 + 0.4842 \times 0) = 24.95$$

The value of the option at node H is therefore

$$0.335 \times 21.14 + 0.665 \times 24.95 = 23.67$$

The value at node G is similarly

$$0.81 \times 102.83 + 0.19 \times 106.64 = 103.56$$

We now move on to the nodes at time 4 years. At node F the option is clearly worth zero. At node E there is a 5% chance that the employee will forfeit the option because he or she leaves the company and a 95% chance that the option will be retained. In the latter case the option is worth

$$e^{-0.05 \times 2}(0.5158 \times 23.67 + 0.4842 \times 0) = 11.05$$

The option is therefore worth $0.95 \times 11.05 = 10.49$. At node D there is a 0.43 probability that the option will be exercised and a 0.57 chance that it will be retained. The value of the option is 56.44.

Consider next the initial node and the nodes at time 2 years. The option has not vested at these nodes. There is a 5% chance that the option will be forfeited and a 95% chance that it will be retained for a further 2 years. This leads to the valuations shown in Figure 15.1. The valuation of the option at the initial node is 14.97. (This compares with a valuation of 17.98 for a regular option using the same tree.)

The Exercise Multiple Approach

Hull and White suggest a simple model where an employee exercises as soon as the option has vested and the ratio of the stock price to the strike price is above a certain level.⁸ They refer to the ratio of stock price to strike price that triggers exercise as the “exercise multiple”. The option can be valued using a binomial or trinomial tree. As outlined in Section 26.6, it is important to construct a binomial or trinomial tree where nodes lie on the stock prices that will lead to exercise. For example, if the strike price is \$30 and the assumption is that employees exercise when the ratio of the stock price to the strike price is 1.5, the tree should be constructed so that there are nodes at a stock price level of \$45. The tree calculations are similar to those for Example 15.2 and take account of the probability of an employee leaving the company.⁹ To estimate the exercise multiple, it is necessary to calculate from historical data the average ratio of

⁸ See J. Hull and A. White, “How to value employee stock options,” *Financial Analysts Journal*, 60, 1 (January/February 2004): 3–9.

⁹ Software implementing this approach is on www.rotman.utoronto.ca/~hull.

stock price to strike price at the time of exercise. (Exercises at maturity and those arising from the termination of the employee's job are not included in the calculation of the average.) This may be easier to estimate from historical data than the expected life because the latter is quite heavily dependent on the particular path that has been followed by the stock's price.

A Market-Based Approach

One way of valuing an employee stock option is to see what the market would pay for it. Cisco was the first to try this in 2006. It proposed selling options with the exact terms of its employee stock options to institutional investors. This approach was rejected by the SEC on the grounds that the range of investors bidding for the options was not wide enough.

Zions Bancorp has suggested an alternative approach. It proposed that securities providing payoffs mirroring those actually realized by its employees be sold. Suppose that the strike price for a particular grant to employees is \$40 and it turns out that 1% of employees exercise after exactly 5 years when the stock price is \$60, 2% exercise after exactly 6 years when the stock price is \$65, and so on. Then 1% of the securities owned by an investor will provide a \$20 payoff after 5 years, 2% will provide a payoff of \$25 after 6 years, and so on.

Zions Bancorp tested the idea using its own stock option grant to its employees. It sold the securities using a Dutch auction process. In this individuals or companies can submit a bid indicating the price they are prepared to pay and the number of options they are prepared to buy. The clearing price is the highest bid such that the aggregate number of options sought at that price or a higher price equals or exceeds the number of options for sale. Buyers who have bid more than the clearing price get their orders filled at the clearing price and the buyer who bid the clearing price gets the remainder. Zions Bancorp announced that it had received SEC approval for its market-based approach in October 2007.

Dilution

The fact that a company issues new stock when an employee stock option is exercised leads to some dilution for existing stock holders because new shares are being sold to employees at below the current stock price. It is natural to assume that this dilution takes place at the time the option is exercised. However, this is not the case. As explained in Section 14.10, stock prices are diluted when the market first hears about a stock option grant. The possible exercise of options is anticipated and immediately reflected in the stock price. This point is emphasized by the example in Business Snapshot 14.3.

The stock price immediately after a grant is announced to the public reflects any dilution. Provided that this stock price is used in the valuation of the option, it is not necessary to adjust the option price for dilution. In many instances the market expects a company to make regular stock option grants and so the market price of the stock anticipates dilution even before the announcement is made.

If a company is contemplating a stock option grant that will surprise the market, the cost can be calculated as described in Example 14.7. This cost can be compared with benefits such as lower regular employee remuneration and less employee turnover.

15.5 BACKDATING SCANDALS

No discussion of employee stock options would be complete without mentioning backdating scandals. Backdating is the practice of marking a document with a date that precedes the current date.

Suppose that a company decides to grant at-the-money options to its employees on April 30 when the stock price is \$50. If the stock price was \$42 on April 3, it is tempting to behave as if those the options were granted on April 3 and use a strike price of \$42. This is legal provided that the company reports the options as \$8 in the money on the date when the decision to grant the options is made, April 30. But it is illegal for the company to report the options as at-the-money and granted on April 3. The value on April 3 of an option with a strike price of \$42 is much less than its value on April 30. Shareholders are misled about the true cost of the decision to grant options if the company reports the options as granted on April 3.

How prevalent is backdating? To answer this question, researchers have investigated whether a company's stock price has, on average, a tendency to be low at the time of the grant date that the company reports. Early research by Yermack shows that stock prices tend to increase after reported grant dates.¹⁰ Lie extended Yermack's work, showing that stock prices also tended to decrease before reported grant dates.¹¹ Furthermore he showed that the pre- and post-grant stock price patterns had become more pronounced over time. His results are summarized in Figure 15.2, which shows average abnormal returns around the grant date for the 1993–94, 1995–98, and 1999–2002 periods. (Abnormal returns are the returns after adjustments for returns on the market portfolio and the beta of the stock.) Standard statistical tests show that it is almost impossible for the patterns shown in Figure 15.2 to be observed by chance. This led both academics and regulators to conclude in 2002 that backdating had become a common practice. In August 2002 the SEC required option grants by public companies to be reported within two business days. Heron and Lie showed that this led to a dramatic reduction in the abnormal returns around the grant dates—particularly for those companies that complied with this requirement.¹² It might be argued that the patterns in Figure 15.2 are explained by managers simply choosing grant dates after bad news or before good news, but the Heron and Lie study provides compelling evidence that this is not the case.

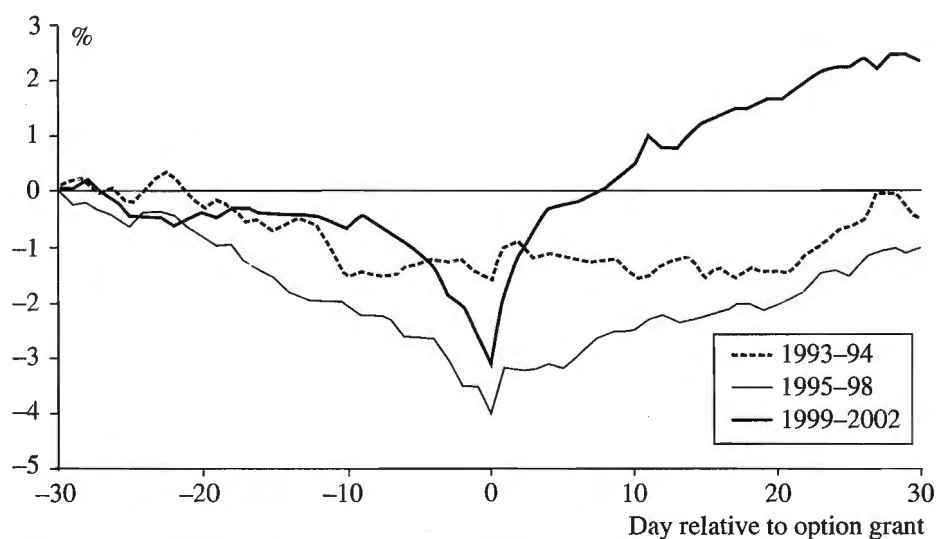
Estimates of the number of companies that illegally backdated stock option grants in the United States vary widely. Tens and maybe hundreds of companies seem to have engaged in the practice. Many companies seem to have adopted the view that it was acceptable to backdate up to one month. Some CEOs resigned when their backdating practices came to light. In August 2007, Gregory Reyes of Brocade Communications Systems, Inc., became the first CEO to be tried for backdating stock option grants. Allegedly, Mr. Reyes said to a human resources employee: "It is not illegal if you do not get caught." In June 2010, he was sentenced to 18 months in prison and fined \$15 million.

¹⁰ See D. Yermack, "Good timing: CEO stock option awards and company news announcements," *Journal of Finance*, 52 (1997), 449–476.

¹¹ See E. Lie, "On the timing of CEO stock option awards," *Management Science*, 51, 5 (May 2005), 802–12.

¹² See R. Heron and E. Lie, "Does backdating explain the stock price pattern around executive stock option grants," *Journal of Financial Economics*, 83, 2 (February 2007), 271–95.

Figure 15.2 Erik Lie's results providing evidence of backdating. (Reproduced with permission, from www.biz.uiowa.edu/faculty/ellie/backdating.htm.)



Companies involved in backdating have had to restate past financial statements and have been defendants in class action suits brought by shareholders who claim to have lost money as a result of backdating. For example, McAfee announced in December 2007 that it would restate earnings between 1995 and 2005 by \$137.4 million. In 2006, it set aside \$13.8 million to cover lawsuits.

SUMMARY

Executive compensation has increased very fast in the last 20 years and much of the increase has come from the exercise of stock options granted to the executives. Until 2005 at-the-money stock option grants were a very attractive form of compensation. They had no impact on the income statement and were very valuable to employees. Accounting standards now require options to be expensed.

There are a number of different approaches to valuing employee stock options. A common approach is to use the Black-Scholes-Merton model with the life of the option set equal to the expected time to exercise or expiry of the option. Another approach is to assume that options are exercised as soon as the ratio of the stock price to the strike price reaches a certain barrier. A third approach is to try and estimate the relationship between the probability of exercise, the ratio of the stock price to the strike price, and the time to option maturity. A fourth approach is to create a market for securities that replicate the payoffs on the options.

Academic research has shown beyond doubt that many companies have engaged in the illegal practice of backdating stock option grants in order to reduce the strike price, while still contending that the options were at the money. The first prosecutions for this illegal practice were in 2007.

FURTHER READING

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Practice Questions (Answers in Solutions Manual)

- 15.1. Why was it attractive for companies to grant at-the-money stock options prior to 2005? What changed in 2005?
- 15.2. What are the main differences between a typical employee stock option and an American call option traded on an exchange or in the over-the-counter market?
- 15.3. Explain why employee stock options on a non-dividend-paying stock are frequently exercised before the end of their lives, whereas an exchange-traded call option on such a stock is never exercised early.
- 15.4. "Stock option grants are good because they motivate executives to act in the best interests of shareholders." Discuss this viewpoint.
- 15.5. "Granting stock options to executives is like allowing a professional footballer to bet on the outcome of games." Discuss this viewpoint.
- 15.6. Why did some companies backdate stock option grants in the US prior to 2002? What changed in 2002?
- 15.7. In what way would the benefits of backdating be reduced if a stock option grant had to be revalued at the end of each quarter?
- 15.8. Explain how you would do the analysis to produce a chart such as the one in Figure 15.2.
- 15.9. On May 31 a company's stock price is \$70. One million shares are outstanding. An executive exercises 100,000 stock options with a strike price of \$50. What is the impact of this on the stock price?
- 15.10. The notes accompanying a company's financial statements say: "Our executive stock options last 10 years and vest after 4 years. We valued the options granted this year using the Black–Scholes–Merton model with an expected life of 5 years and a volatility of 20%." What does this mean? Discuss the modeling approach used by the company.

- 15.11. In a Dutch auction of 10,000 options, bids are as follows:
 A bids \$30 for 3,000
 B bids \$33 for 2,500
 C bids \$29 for 5,000
 D bids \$40 for 1,000
 E bids \$22 for 8,000
 F bids \$35 for 6,000.
 What is the result of the auction? Who buys how many at what price?
- 15.12. A company has granted 500,000 options to its executives. The stock price and strike price are both \$40. The options last for 12 years and vest after 4 years. The company decides to value the options using an expected life of 5 years and a volatility of 30% per annum. The company pays no dividends and the risk-free rate is 4%. What will the company report as an expense for the options on its income statement?
- 15.13. A company's CFO says: "The accounting treatment of stock options is crazy. We granted 10,000,000 at-the-money stock options to our employees last year when the stock price was \$30. We estimated the value of each option on the grant date to be \$5. At our year-end the stock price had fallen to \$4, but we were still stuck with a \$50 million charge to the P&L." Discuss.

Further Questions

- 15.14. What is the (risk-neutral) expected life for the employee stock option in Example 15.2? What is the value of the option obtained by using this expected life in Black-Scholes-Merton?
- 15.15. A company has granted 2,000,000 options to its employees. The stock price and strike price are both \$60. The options last for 8 years and vest after 2 years. The company decides to value the options using an expected life of 6 years and a volatility of 22% per annum. Dividends on the stock are \$1 per year, payable halfway through each year, and the risk-free rate is 5%. What will the company report as an expense for the options on its income statement?
- 15.16. A company has granted 1,000,000 options to its employees. The stock price and strike price are both \$20. The options last 10 years and vest after 3 years. The stock price volatility is 30%, the risk-free rate is 5%, and the company pays no dividends. Use a four-step tree to value the options. Assume that there is a probability of 4% that an employee leaves the company at the end of each of the time steps on your tree. Assume also that the probability of voluntary early exercise at a node, conditional on no prior exercise, when (a) the option has vested and (b) the option is in the money, is

$$1 - \exp[-a(S/K - 1)/T]$$

where S is the stock price, K is the strike price, T is the time to maturity, and $a = 2$.